

TAO Yuhao

EDUCATION

ETH Zürich, Switzerland

M.Sc. in Robotics, Systems and Control (with distinction)

Sept. 2023-present

Grade: 5.84/6.0

- **Core Courses:** Microrobotics, Nanorobotics, Introduction to Machine Learning, Image Analysis and Computer Vision, Dynamic Programming and Optimal Control, Linear System Theory, Recursive Estimation, Biomedical Imaging.

RWTH Aachen University, Germany

B.Sc. in Mechanical Engineering (Graduation with distinction)

Oct. 2018-Jul. 2023

Grade: 1.1/1.0 (Ranking: 2/1885)

- **Awards:** Dean's List 2019/2020, 2020/2021 (Top 5%); Germany Scholarship 2020/2021, 2022/2023
- **Core Courses:** Mechanics, Material Science, Machine Design, Simulation Methods in Engineering, Automatic Control, Numerical Mathematics, Fluid Mechanics, Thermodynamics, etc.
- **GRE on-site:** 336 (Verbal 166, Quantitative 170) + 3.5 (Writing); **TOEFL on-site:** 112 (Reading 30, Listening 28, Speaking 28, Writing 26)

RESEARCH EXPERIENCE

Design and Fabrication of An Smart Millirobot with On-board Sensing and Control

Apr. 2024-Sept. 2025

(Master Thesis & Semester Project @ Multi-Scale Robotics Lab)

ETH Zürich

Supervisor: Dr. Bradley Nelson, Professor / Advisor: Quan Gao, Ph.D. Candidate

- Designed, fabricated and optimized the umbrella structure of the robot based on Pop-up MEMS fabrication technique
- Designed, optimized and characterized the 3D antenna and integrated it onto the robot structure
- Designed the on-board PCB for control and sensing, and implemented the code for MCU
- Developed a simulation platform based on ABAQUS using Python and Fortran for simulating interactions between mobile cube magnets

Design and Fabrication of An Untethered Adhesive Patch Robot with On-board Power and Sensing

Sept. 2024- Sept. 2025

(Master Thesis @ Multi-Scale Robotics Lab)

ETH Zürich

Supervisor: Dr. Bradley Nelson, Professor / Advisor: Quan Gao, Ph.D. Candidate

- Optimized and calibrated the on-board PCB for low power consumption, and precise voltage measurement and data transmission
- Fabricated the adhesive patches based on different recipes and compared their performance using peeling force test
- Fabricated the magnetic PDMS layer and integrated it with the adhesive patch

Modeling and simulation of deformable Kolmogorov-sized bubbles and droplets in turbulent flows

Jan. 2023-Apr. 2023

(Bachelor Thesis @ Institute of Aerodynamics)

RWTH Aachen University

Supervisor: Dr. Wolfgang Schröder, Professor / Advisor: Julian Vorspohl, Ph.D. Candidate

- Extended the Lagrangian Particle Tracking (LPT) solver of the in-house flow solver in AIA using C++ to account for particle deformability
- Tested the implemented module in relaxation, shear flow, vortex, and rising bubble test cases
- Studied the influence of the density ratio, particle size, and deformability on bubble/droplet-laden decaying isotropic turbulence flow

Atomistic Simulations of Tensile Deformation in A CrCoNi Medium-Entropy Alloy

Dec. 2021-Nov. 2022

(External Research Project @ State Key Laboratory of Nonlinear Mechanics)

Institute of Mechanics, CAS

Supervisor: Dr. Xiaolei Wu, Professor / Advisor: Wenqiang Cheng, Ph.D. Candidate

- Designed the simulation process and modeled the initial atom arrangements using the Voronoi method on MATLAB
- Conducted the molecular dynamics simulations with LAMMPS
- Developed algorithms to investigate the structural and strain features of simulation results utilizing OVITO and MATLAB

Simulation of Electric Current Assisted Sintering of Oxide Ceramics

Nov. 2021-May 2022

(Semester Project @ Institute for Materials Applications in Mechanical Engineering)

RWTH Aachen University

Supervisor: Dr. Christoph Broeckmann, Professor / Advisor: Shufan Wang, Ph.D. Candidate

- Simulated the Flash sintering process on MATLAB via Skorohold-Olevsky Viscous Sintering (SOVS) model
- Simulated the UHS- and Flash sintering process using finite element method on ABAQUS with user subroutine of SOVS model
- Applied the Parameter Calibration on MATLAB using damped Newton's method

Phase-Field Simulation of Sintering Process

Sept. 2020-Feb. 2021

(Undergraduate Research Assistant @ Institute for Materials Applications in Mechanical Engineering)

RWTH Aachen University

Supervisor: Dr. Christoph Broeckmann, Professor / Advisor: Shufan Wang, Ph.D. Candidate

- Built a distributed-parameter Phase Field model considering grain orientation and rigid body motion on MATLAB
- Simulated the sintering process on MATLAB utilizing the developed Phase-Field model and on MOOSE

PUBLICATION

Tao Y, Cheng W and Wang W (2023) Atomistic tensile deformation mechanisms in a CrCoNi medium-entropy alloy with gradient nano-grained structure at cryogenic temperature. *Front. Mater.* 9:1118952. doi: 10.3389/fmats.2022.1118952

INTERNSHIP

Mercedes-Benz Group | Research & Development Intern | Untertürkheim, Stuttgart, Germany

Jul. 2022-Jan. 2023

- Developed numerical algorithms for 2D~3D scattering points interpolation on MATLAB to improve the battery cell models
- Optimized the distribution of test points utilizing the Monte Carlo method to minimize the interpolation error of the battery cell model
- Developed graphical user interfaces (GUI) on MATLAB for visualization and post-processing of experimental data of battery cells

SKILLS

Language: Chinese (Native); English (Fluent); German (Fluent); Russian (Elementary)

Software: Siemens NX, MATLAB, ABAQUS, Ansys HFSS, LAMMPS, MOOSE, OVITO, Origin, Blender, Keil, CubeMX

Programming language: C++, Python, MATLAB, Fortran, Java